Systems & Inequalities

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Geometric sequences as exponential functions

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 16 cm. Multiplying the original length by the scale factor gives $4 \times 4 = 16$ cm.

2 [Foundational] • Target: Geometric sequences as exponential functions

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

Final Answer: 16. Substituting $x = 3$ into $y = 4x + 4$ gives $y = 4(3) + 4 = 16$.

3 [Foundational] • Target: Geometric sequences as exponential functions

In ΔABC studying geometric sequences as exponential functions, $m\angle A = 30^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 84°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (30^\circ + 66^\circ) = 84^\circ$.

4 [Foundational] • Target: Geometric sequences as exponential functions

A geometric figure scaled by factor $k = 5$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 20 cm. Multiplying the original length by the scale factor gives $4 \times 5 = 20$ cm.

5 [Foundational] • Target: Geometric sequences as exponential functions

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

Final Answer: 20. Substituting $x = 4$ into $y = 4x + 4$ gives $y = 4(4) + 4 = 20$.

6 [Foundational] • Target: Comparing exponential relationships

For the model $f(x) = 3x + 5$ in comparing exponential relationships, evaluate $f(2)$.

Final Answer: 11. Substitute $x = 2$: $f(2) = 3(2) + 5 = 11$.

7 [Foundational] • Target: Comparing exponential relationships

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

Final Answer: $x = 1$ and $x = 6$. Factoring $(x - 1)(x - 6) = 0$ yields the roots $x = 1$ and $x = 6$.

© ClipSAT — clipsat.org | All Rights Reserved

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Comparing exponential relationships

A data set recorded for comparing exponential relationships consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 22.5. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

9 [Intermediate] • Target: Comparing exponential relationships

For the model $f(x) = 3x + 5$ in comparing exponential relationships, evaluate $f(3)$.

Final Answer: 14. Substitute $x = 3$: $f(3) = 3(3) + 5 = 14$.

10 [Intermediate] • Target: Comparing exponential relationships

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

Final Answer: $x = 2$ and $x = 7$. Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

11 [Intermediate] • Target: Comparing exponential relationships

For the model $f(x) = 3x + 5$ in comparing exponential relationships, evaluate $f(4)$.

Final Answer: 17. Substitute $x = 4$: $f(4) = 3(4) + 5 = 17$.

12 [Intermediate] • Target: Comparing exponential relationships

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

Final Answer: $x = 3$ and $x = 8$. Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

13 [Intermediate] • Target: Comparing exponential relationships

For the model $f(x) = 3x + 5$ in comparing exponential relationships, evaluate $f(5)$.

Final Answer: 20. Substitute $x = 5$: $f(5) = 3(5) + 5 = 20$.

14 [Intermediate] • Target: Comparing exponential relationships

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

Final Answer: $x = 4$ and $x = 9$. Factoring $(x - 4)(x - 9) = 0$ yields the roots $x = 4$ and $x = 9$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

© ClipSAT — clipsat.org | All Rights Reserved

15 [Challenging] • Target: Comparing linear and exponential relationships • Cross-Lesson: 10.01 Rational Expressions

For the model $f(x) = 4x + 6$ in comparing linear and exponential relationships, evaluate $f(2)$.

Final Answer: 14. Substitute $x = 2$: $f(2) = 4(2) + 6 = 14$.

16 [Challenging] • Target: Comparing linear and exponential relationships • Cross-Lesson: 11.01 Statistics & Data Analysis

A data set recorded for comparing linear and exponential relationships consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 23.5. The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

17 [Challenging] • Target: Comparing linear and exponential relationships • Cross-Lesson: 2.01 Quadratic Functions

For the model $f(x) = 4x + 6$ in comparing linear and exponential relationships, evaluate $f(3)$.

Final Answer: 18. Substitute $x = 3$: $f(3) = 4(3) + 6 = 18$.

18 [Challenging] • Target: Comparing linear and exponential relationships • Cross-Lesson: 3.01 Exponents & Polynomials

For the model $f(x) = 4x + 6$ in comparing linear and exponential relationships, evaluate $f(4)$.

Final Answer: 22. Substitute $x = 4$: $f(4) = 4(4) + 6 = 22$.

19 [Challenging] • Target: Comparing linear and exponential relationships • Cross-Lesson: 4.01 Functions & Transformations

For the model $f(x) = 4x + 6$ in comparing linear and exponential relationships, evaluate $f(5)$.

Final Answer: 26. Substitute $x = 5$: $f(5) = 4(5) + 6 = 26$.

20 [Challenging] • Target: Simple interest • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

For the model $f(x) = 5x + 7$ in simple interest, evaluate $f(2)$.

Final Answer: 17. Substitute $x = 2$: $f(2) = 5(2) + 7 = 17$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Graphs and characteristics of exponential functions

For the model $f(x) = 5x + 9$ in graphs and characteristics of exponential functions, evaluate $f(2)$.

a. 19 ✓

b. 24

c. 10

d. 23

Substitute $x = 2$: $f(2) = 5(2) + 9 = 19$.

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Graphs and characteristics of exponential functions

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

23 [Foundational] • Target: Graphs and characteristics of exponential functions

For the model $f(x) = 5x + 9$ in graphs and characteristics of exponential functions, evaluate $f(5)$.

a. 34 ✓

b. 39

c. 25

d. 38

Substitute $x = 5$: $f(5) = 5(5) + 9 = 34$.

24 [Foundational] • Target: Exponential growth and decay

For the model $f(x) = 6x + 3$ in exponential growth and decay, evaluate $f(2)$.

a. 15 ✓

b. 21

c. 12

d. 19

Substitute $x = 2$: $f(2) = 6(2) + 3 = 15$.

25 [Foundational] • Target: Exponential growth and decay

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

26 [Foundational] • Target: Exponential growth and decay

For the model $f(x) = 6x + 3$ in exponential growth and decay, evaluate $f(5)$.

a. 33 ✓

b. 39

c. 30

d. 37

Substitute $x = 5$: $f(5) = 6(5) + 3 = 33$.

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Geometric sequences as exponential functions

In ΔABC studying geometric sequences as exponential functions, $m\angle A = 30^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 105° ✓

b. 115°

c. 97°

d. 150°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (30^\circ + 45^\circ) = 105^\circ$.

28 [Foundational] • Target: Geometric sequences as exponential functions

A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length?

a. 9 cm

b. 12 cm ✓

c. 16 cm

d. 36 cm

Multiplying the original length by the scale factor gives $4 \times 3 = 12$ cm.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Graphs and characteristics of exponential functions

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ **and** $x = 3$ ✓

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

30 [Intermediate] • Target: Graphs and characteristics of exponential functions

A data set recorded for graphs and characteristics of exponential functions consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

© ClipSAT — clipsat.org | All Rights Reserved

31 [Intermediate] • Target: Graphs and characteristics of exponential functions

For the model $f(x) = 5x + 9$ in graphs and characteristics of exponential functions, evaluate $f(3)$.

a. 24 ✓

b. 29

c. 15

d. 28

Substitute $x = 3$: $f(3) = 5(3) + 9 = 24$.

32 [Intermediate] • Target: Graphs and characteristics of exponential functions

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ **and** $x = 5$ ✓

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

33 [Intermediate] • Target: Graphs and characteristics of exponential functions

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ **and** $x = 6$ ✓

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

34 [Intermediate] • Target: Exponential growth and decay

A data set recorded for exponential growth and decay consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

© ClipSAT — clipsat.org | All Rights Reserved

35 [Intermediate] • Target: Exponential growth and decay

For the model $f(x) = 6x + 3$ in exponential growth and decay, evaluate $f(3)$.

a. 21 ✓

b. 27

c. 18

d. 25

Substitute $x = 3$: $f(3) = 6(3) + 3 = 21$.

36 [Intermediate] • Target: Exponential growth and decay

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ **and** $x = 6$ ✓

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

37 [Intermediate] • Target: Exponential growth and decay

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ **and** $x = 7$ ✓

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

Factoring $(x - 4)(x - 7) = 0$ yields the roots $x = 4$ and $x = 7$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Geometric sequences as exponential functions • Cross-Lesson: 3.01 Exponents & Polynomials

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 1$?

a. 4

b. 10

c. 8 ✓

d. 12

Substituting $x = 1$ into $y = 4x + 4$ gives $y = 4(1) + 4 = 8$.

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Geometric sequences as exponential functions • Cross-Lesson: 4.01 Functions & Transformations

In ΔABC studying geometric sequences as exponential functions, $m\angle A = 30^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 98° ✓

b. 108°

c. 90°

d. 150°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (30^\circ + 52^\circ) = 98^\circ$.

40 [Challenging] • Target: Geometric sequences as exponential functions • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

In ΔABC studying geometric sequences as exponential functions, $m\angle A = 30^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 91° ✓

b. 101°

c. 83°

d. 150°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (30^\circ + 59^\circ) = 91^\circ$.

41 [Challenging] • Target: Geometric sequences as exponential functions • Cross-Lesson: 7.01 Sequences & Series

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm ✓

c. 20 cm

d. 64 cm

Multiplying the original length by the scale factor gives $4 \times 4 = 16$ cm.

42 [Challenging] • Target: Geometric sequences as exponential functions • Cross-Lesson: 8.01 Factoring Polynomials

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$?

a. 16

b. 22

c. 20 ✓

d. 24

Substituting $x = 4$ into $y = 4x + 4$ gives $y = 4(4) + 4 = 20$.

© ClipSAT — clipsat.org | All Rights Reserved

43 [Challenging] • Target: Comparing exponential relationships • Cross-Lesson: 9.01 Radicals & Exponents

A data set recorded for comparing exponential relationships consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5 ✓

d. 24.5

The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

44 [Challenging] • Target: Simple interest • Cross-Lesson: 1.01 Linear Equations & Lines

A data set recorded for simple interest consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

45 [Challenging] • Target: Simple interest • Cross-Lesson: 10.01 Rational Expressions

For the model $f(x) = 5x + 7$ in simple interest, evaluate $f(3)$.

a. 22 ✓

b. 27

c. 15

d. 26

Substitute $x = 3$: $f(3) = 5(3) + 7 = 22$.

© ClipSAT — clipsat.org | All Rights Reserved